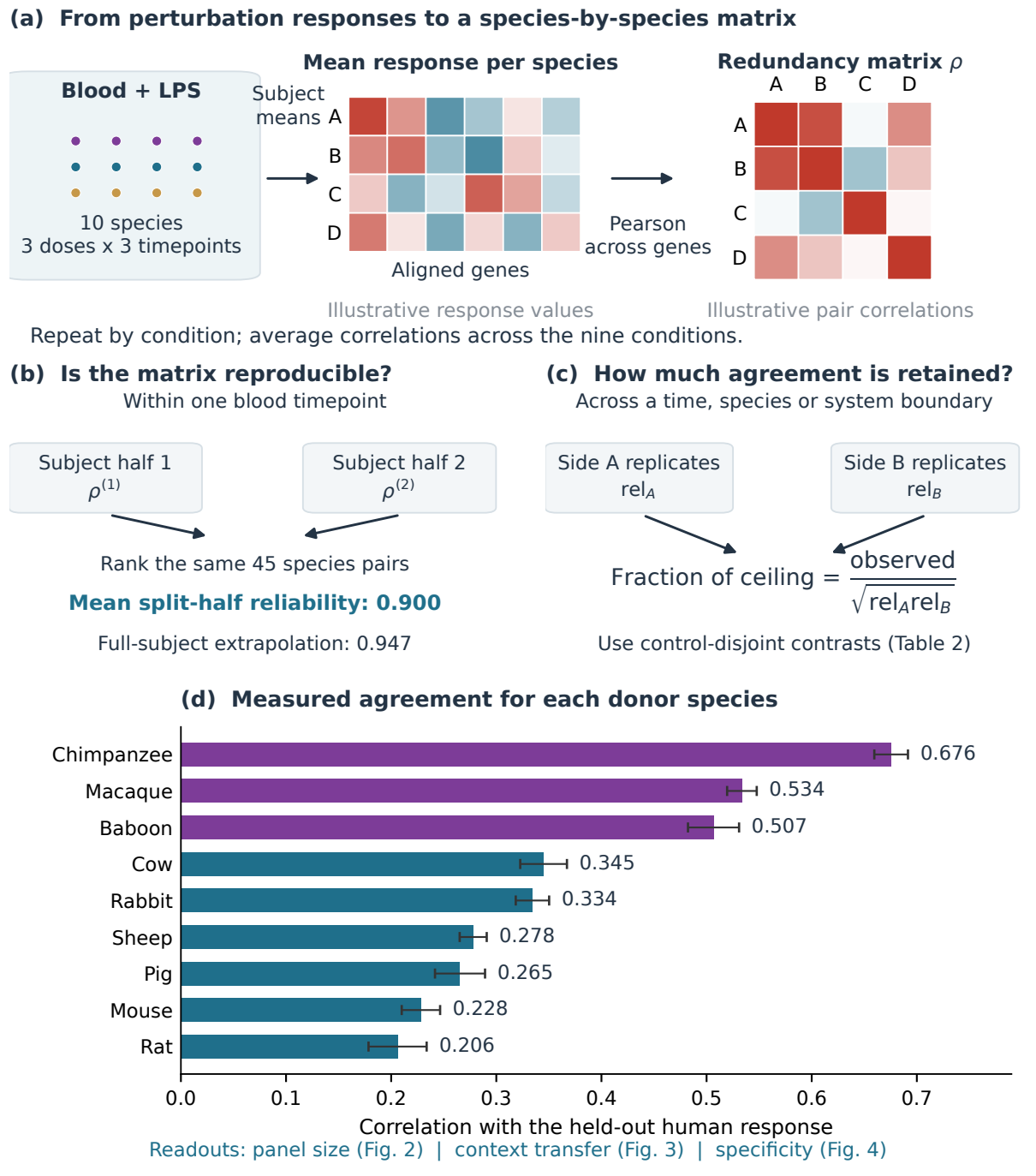
(a) From perturbation responses to a species-by-species matrix
Blood + LPS
10 species
3 doses x 3 timepoints
Mean response per species
Subject means
A
B
C
D
Aligned genes
Pearson across genes
Redundancy matrix ρ
A B C D
A
B
C
D
Illustrative response values
Illustrative pair correlations
Repeat by condition; average correlations across the nine conditions.
(b) Is the matrix reproducible?
Within one blood timepoint
Subject half 1 ρ(1)
Subject half 2 ρ(2)
Rank the same 45 species pairs
Mean split-half reliability: 0.900
Full-subject extrapolation: 0.947
(c) How much agreement is retained?
Across a time, species or system boundary
Side A replicates rel_A
Side B replicates rel_B
Fraction of ceiling = observed / √(rel_A rel_B)
Use control-disjoint contrasts (Table 2)
(d) Measured agreement for each donor species
Chimpanzee 0.676
Macaque 0.534
Baboon 0.507
Cow 0.345
Rabbit 0.334
Sheep 0.278
Pig 0.265
Mouse 0.228
Rat 0.206
0.0 0.1 0.2 0.3 0.4 0.5 0.6 0.7
Correlation with the held-out human response
Readouts: panel size (Fig. 2) | context transfer (Fig. 3) | specificity (Fig. 4)